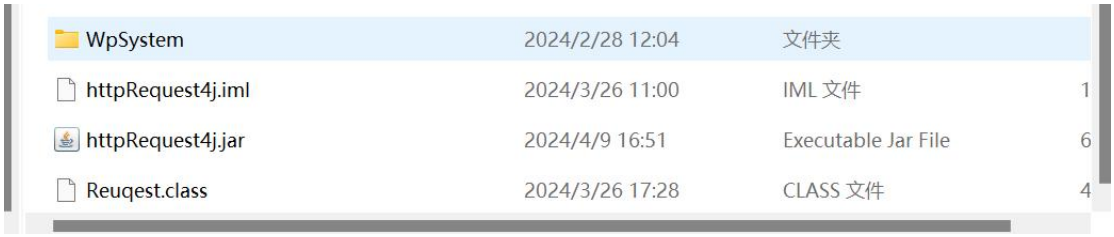
实验 Java5

先把 jar 包下载下来



WpSystem	2024/2/28 12:04	文件夹	
httpRequest4j.iml	2024/3/26 11:00	IML 文件	1
httpRequest4j.jar	2024/4/9 16:51	Executable Jar File	6
Reuqest.class	2024/3/26 17:28	CLASS 文件	4

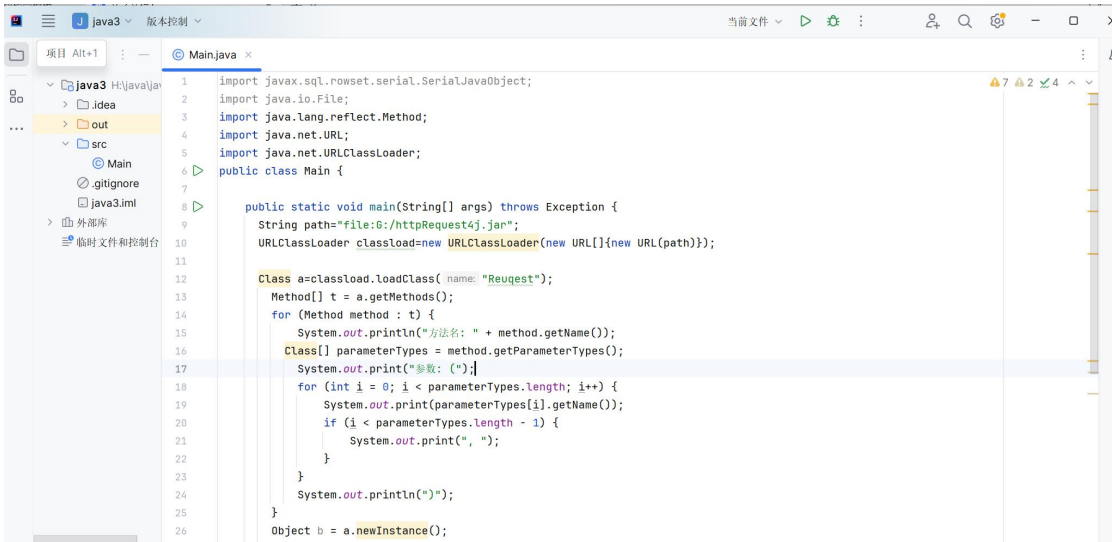
使用命令行查看包里有什么类，方便我们去通过反射去查看类中的方法

```
G:\>jar tf httpRequest4j.jar
META-INF/MANIFEST.MF
.idea/
.idea/.gitignore
.idea/artifacts/
.idea/artifacts/httpRequest4j_jar.xml
.idea/misc.xml
.idea/modules.xml
.idea/workspace.xml
httpRequest4j.iml
META-INF/
Reuqest.class

G:\>
```

看到有一个 Reuqest(应该是 request？)的类

编写代码



```java
import javax.sql.rowset.serial.SerialJavaObject;
import java.io.File;
import java.lang.reflect.Method;
import java.net.URL;
import java.net.URLClassLoader;
public class Main {

    public static void main(String[] args) throws Exception {
      String path="file:G:/httpRequest4j.jar";
      URLClassLoader classload=new URLClassLoader(new URL[]{new URL(path)});

      Class a=classload.loadClass( name: "Reuqest");
        Method[] t = a.getMethods();
        for (Method method : t) {
            System.out.println("方法名: " + method.getName());
          Class[] parameterTypes = method.getParameterTypes();
            System.out.print("参数: ");
            for (int i = 0; i < parameterTypes.length; i++) {
                System.out.print(parameterTypes[i].getName());
                if (i < parameterTypes.length - 1) {
                    System.out.print(", ");
                }
            }
            System.out.println(")");
        }
        Object b = a.newInstance();
```

String path="file:G:/httpRequest4j.jar";

URLClassLoader classload=new URLClassLoader(new URL[]{new URL(path)});

将 jar 包加载进程序

Class a=classload.loadClass("Reuqest");

Method[] t = a.getMethods();

获得一个 class 类

并获取它里面的方法

Getmethons 只能获取公有类型的方法

循环打印方法和参数

```
运行  Main
F:\jdk17\bin\java.exe "-javaagent:F:\ij\IntelliJ IDEA Community Edition 2023.3.6\lib\idea_rt.jar=61258:F:\ij\IntelliJ IDEA Community Edition 2023.3.6\bin" -D
方法名: main
参数: ([Ljava.lang.String;)
方法名: wait
参数: (long, int)
方法名: wait
参数: ()
方法名: wait
参数: (long)
方法名: equals
参数: (java.lang.Object)
方法名: toString
参数: ()
方法名: hashCode
参数: ()
方法名: getClass
参数: ()
方法名: notify
参数: ()
方法名: notifyAll
参数: ()

进程已结束，退出代码为 0
```

通过 getDeclaredMethods 方法获取私有的方法

```
运行  Main
F:\jdk17\bin\java.exe "-javaagent:F:\ij\IntelliJ IDEA Community Edition 2023.3.6\lib\idea_rt.jar=61996:F:\ij\IntelliJ IDEA Community Edition 2023.3.6\bin" -D
方法名: main
参数: ([Ljava.lang.String;)

进程已结束，退出代码为 0
```